MIL-C-3383A (MU)
28 June 1965
SUPERSEDING
MIL-C-3383
25 January 1951

MILITARY SPECIFICATION
CUPS, BULLET JACKET, COPPER ALLOY, FOR SMALL ARMS AMMUNITION

1. SCOPE

1.1 Scope.- This specification covers annealed copper alloy bullet jacket cups used in the manufacture of bullets for small arms cartridges.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION
Military
MIL-C-3993 - Copper and Copper-Base Alloy Mill Products; Packaging Of

STANDARDS
Federal
Federal Test Method Standard No. 151 - Metals; Test Methods

Military
MIL-STD- 105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD- 109 - Quality Assurance Terms and Definitions
MIL-STD- 129 - Marking for Shipment and Storage
MIL-STD-1232 - Visual Inspection Standards for Cups and Disks used in Small Arms Ammunition Manufacture

FSC 1305

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2.1 (Cont)

DRAWINGS

All bullet jacket cup drawings specified in the contract or purchase order (see 6.1).

U. S. Army Munitions Command

D7553671 - Box, Wirebound, Shipping for Small Arms Ammunition Cups and Disks for Bullet Jackets and Cases - List of Parts and Assembly

F7553844 - Box, Shipping for Small Arms Ammunition Cups and Disks for Bullet Jackets and Cases - List of Parts and Assembly

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- The bullet jacket cup shall comply with the requirements specified on the applicable drawing and referenced specifications, and shall process satisfactorily into a bullet jacket.

3.2 Initial production sample.- Unless otherwise directed by the contracting officer, an initial production sample shall be required.

3.3 Grain size.- The grain size on the sidewall of the cup shall be 0.025 to 0.050 mm.

3.4 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.4.1 Metal defects.- The cup shall be free of laminations, slivers, laps, cracks, scale, scratches, dents, bent sidewalls, V'd edges and wire edges.

3.4.2 Foreign matter.- The cup shall be clean and free of oil, grease, dirt, grit, oxidation, stains and chips.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 for definition of quality assurance terms used.

4.2 Initial production sample.- At the beginning of regular production, the supplier shall submit an initial production sample to a Government approved facility for evaluation. The sample shall consist of 5200 cups. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production. All parts and materials must be the same as used in regular production and shall be obtained from the same source of supply. After inspection and provisional acceptance at source, 200 cups of the initial production sample shall be inspected at the Government approved facility for all requirements of the drawings and specifications, and the remaining 5000 cups shall be used for performance of the work test. The Government reserves the right to require new initial production samples until such time as an acceptable sample is submitted.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

4.3.2 Examination.- Examination for major and minor defects shall be performed on a class basis in accordance with the classification of defects, Table I, using applicable sampling plans and acceptance criteria of MIL-STD-105. All nonconforming cups shall be rejected. The acceptable quality levels (AQL) shall be as follows:

Defect Class	AQL (Percent)
Major Visual	0.25
Minor Visual	2.50
Major Dimensional	0.25
Major Weight	0.25

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4.3.2.1 Classification of defects.- The classification of defects shall be as specified in Table I.

Table I

Defect		Method of Inspection
MAJOR		
101.	Scaly metal	Visual
102.	Improperly formed or mutilated	Visual
103.	Laminations or laps	Visual
104.	Outside diameter	Gage
105.	Base thickness	Gage
106.	Wall thickness	Gage
107.	Wall thickness variation	Gage
108.	Wall height variation	Gage
109.	Weight	Scale
MINOR		
201.	Oil, grease, dirt	Visual
202.	Dents or bent edges	Visual
203.	Oxidation	Visual
204.	Scratches	Visual

NOTE: Refer to applicable gilding metal cup section of MIL-STD-1232 for visual defect standards for visual defects listed in Table I.

4.3.3 Tests.- The following tests shall be conducted in accordance with methods and procedures specified in 4.4.

4.3.3.1 Grain size.- A sample of 15 cups shall be tested for grain size. Failure of 2 or more cups to comply with the applicable requirements of 3.3 shall be cause for rejection of the lot. If one cup fails in the first test, a second sample consisting of double the number of cups in the first sample shall be tested. If any cups in the second sample fail to comply with applicable requirements, the lot shall be rejected.

4.3.3.2 Workability.- A sample of 5000 cups from the initial production sample shall be subjected to a work test to determine compliance with the processing requirement specified in 3.1.

4.3.4 Packaging, packing and marking.- Inspection for packaging, packing and marking shall be as directed by the contracting officer.

4.3.5 Inspection equipment.- The examinations shall be made using commercial measuring equipment with a precision of 10 percent of the tolerance on the product dimension.

4.4 Test method and procedures.

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4.4.1 Grain size.- The test shall be conducted in accordance with Method 312 in Federal Test Method Standard No. 151. Grain size measurements shall be made in a zone approximately midway of the length of the sidewall and in the location with respect to the wall thickness of the cup as shown in figure 1. At least 3 readings shall be made, averaged and recorded for each grain size determination.

4.4.2 Workability.- The work test shall comprise all of the process production operations used in the manufacture of the bullet jacket.

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 Level A (Worldwide shipment).- The cups shall be packed in accordance with MIL-C-3993. Gross weight of each pallet type box with cups shall not exceed 2000 pounds.

5.1.2 Level C (Domestic shipment).- The cups shall be packed in pallet boxes conforming to Drawings D7553671 and F7553844. Gross weight of each pallet type box with cups shall not exceed 3000 pounds. Only one type box shall be used for any one lot of cups.

5.2 Marking.- Each box shall be marked with the name of the material, size and type of cup, lot number, name of contractor, and the number of the contract or order in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data.- Invitations for bids and contracts or orders will specify the following:

6.1.1 Title, number and date of this specification.

6.1.2 Drawing number of cup.

6.1.3 Level of packing.

6.1.4 Provision for submission of acceptance inspection reports, in duplicate, to the contracting officer, containing final inspection results for each lot of cups presented to the Government.

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6.1.5 Provisions for performance of work test.- Where feasible, the work test portion of the initial production sample should be forwarded to the facility designated to receive the procured cups. That facility should perform the work test using tools, drawing and cleaning compounds, heat treating processes, lubricants and equipment in accordance with that facility's manufacturing process. The remainder of the initial production sample should be forwarded to a Government approved facility for inspection. The work test should not be performed until acceptability of the cups has been determined in the inspection phase and instructions have been issued by the Government to proceed with the test. In those instances where cups are being procured for delivery to multiple consignees, provisions of 4.2 should apply.

6.2 The supplier shall certify that each lot of cups is coated with a residual film of neutral soap and that a dichromate solution was not used during the pickling operation.

Custodian:
Army - MU

Preparing activity:
Army - MU
Project No. 1305-A348

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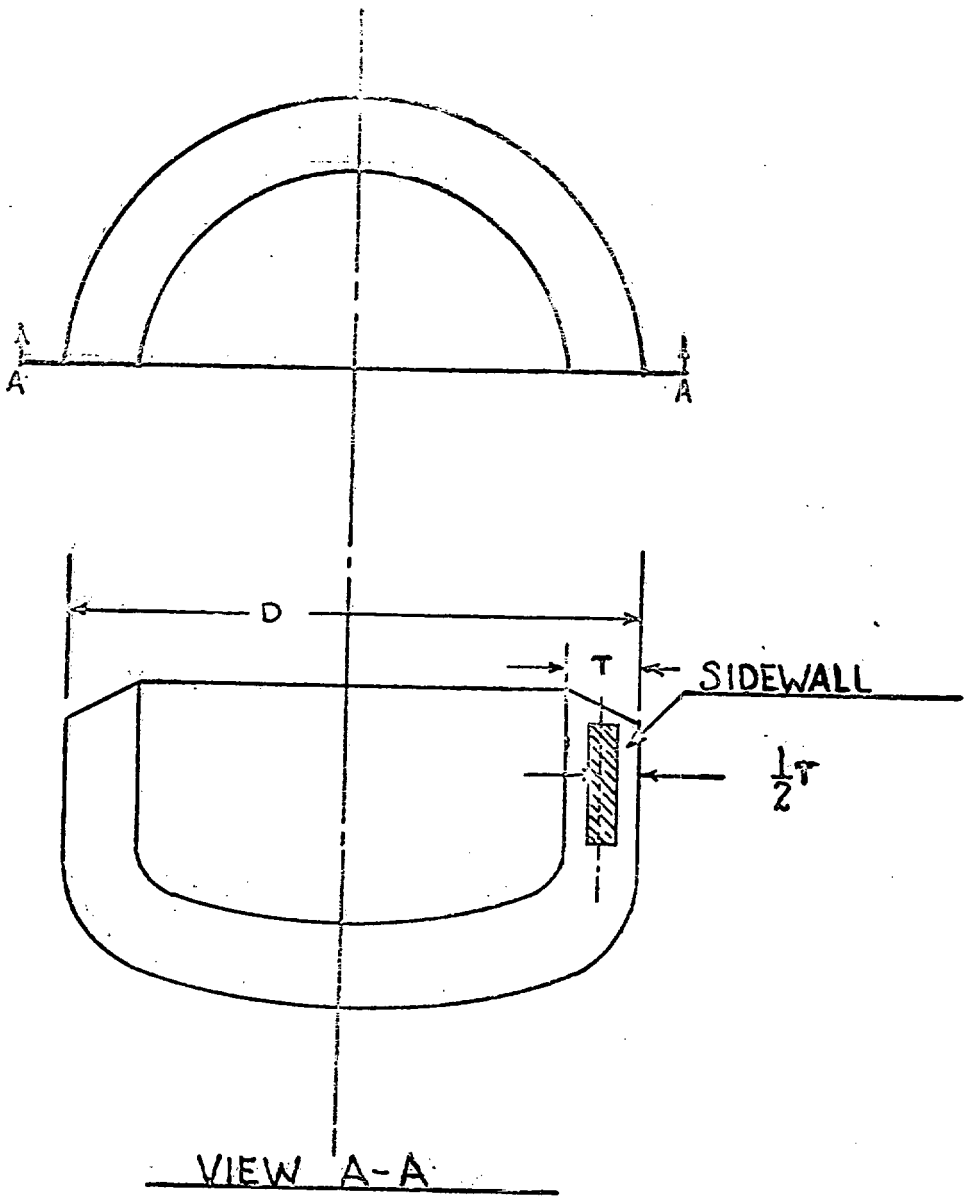


FIGURE 1.—*Zone to be examined for grain size*

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE	
CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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